

Receiver-Guided Reinforcement Learning-based Congestion Control for IoT

Qihang Jiao, *Student Member, IEEE*, and Lotfi Mhamdi, *Senior Member, IEEE*

Abstract—The abstract goes here.

Index Terms—IoT, TCP, congestion control, reinforcement learning.

I. INTRODUCTION

THIS part is to be completed.

II. RELATED WORKS

III. SYSTEM DESIGN

Our proposal contains three modules in the receiver side for adjusting sender CWND, Receiver Measurement Module, Window Decision Module, and Window Enforcement Module. Additionally, TCP sockets in both sender and receiver are extended or modified for adding interfaces to achieved our features.

A. Message types definition

The receiver TCP socket is extended to support communication with the RL agent. Three types of messages are defined:

- **State**: carries the observed network state and is sent to the agent for storage in the replay buffer.
- **Action Request**: triggers the agent to infer and return a control action based on the current state.
- **Connection Close**: notifies the agent that a flow has terminated, allowing it to clear the corresponding internal states.

B. Receiver Measurement Module

Our scheme relies on reliable state observations for a stable training. For a receiver, RTT is not able to be directly observed while it is valuable metric to measure the congestion extent of a network. We overcome this by extending the sender TCP socket to include their CWND in TCP header. In standard TCP congestion avoidance, CC increases increase by approximately 1 MSS after a RTT, i.e., when an entire CWND bytes are acknowledged. We utilise this to enable the receiver to estimate RTT by calculating the elapsed time for a flow to complete its CWND. Therefore, upon every packet receive event that includes a CWND value different from the last value received, the Receiver Measurement Modules establishes a new estimation task. A new task is initialised with the current

Qihang Jiao and Lotfi Mhamdi are with the School of Electronic and Electrical Engineering, University of Leeds, Leeds, UK (e-mail: el22qj@leeds.ac.uk; L.Mhamdi@leeds.ac.uk).

Lotfi Mhamdi is also with the College of Computing & Systems, Abdullah Al Salem University, Kuwait (e-mail: lotfi.mhamdi@asu.edu.kw).

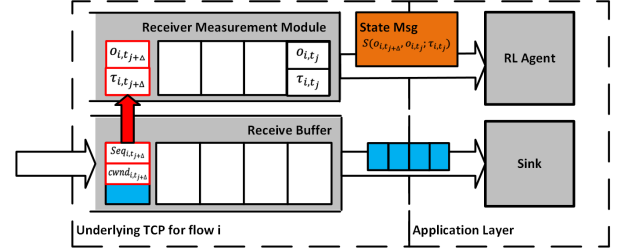


Fig. 1. An example of estimation task completion of flow i .

observations and a threshold, e.g., for flow i at time j , defined as below:

$$\tau_{i,t_j} = Seq_{i,t_j} + cwnd_{i,j}$$

An estimation task is completed when a coming packet with $Seq_{i,t_{j+\Delta}} > \tau_{i,t_j}$, the $cwnd$ elapsed time is calculated as $t_{elapsed} = t_{i,j+\Delta} - t_{i,j}$. Then $t_{elapsed}$ along with other observations ($o_{i,t_{j+\Delta}}, o_{i,t_j}$) are processed to generate a state for training our RL agent.

C. Window Decision Module

The states input is triggered by each receive event that is not able to be known when it happens. Also for the purpose to avoid the agent decision process blocking the normal TCP behaviour, we set a window decision maintained in the Receiver socket and updated periodically. The window decision is a value within $[0, 1]$ and set in every ACK header to the sender.

D. Window Enforcement Module

The sender is modified to leverage the window decision as a factor to update its CWND as following:

$$CWND \leftarrow CWND \times Scaling_Factor^{Decision}$$

E. Enforcing Window Generation by RL Agent

IV. SIMULATION SETUP AND RESULTS

V. CONCLUSION

The conclusion goes here.

ACKNOWLEDGMENT

The authors would like to thank...

REFERENCES

- [1] H. Kopka and P. W. Daly, *A Guide to LATEX*, 3rd ed. Harlow, England: Addison-Wesley, 1999.